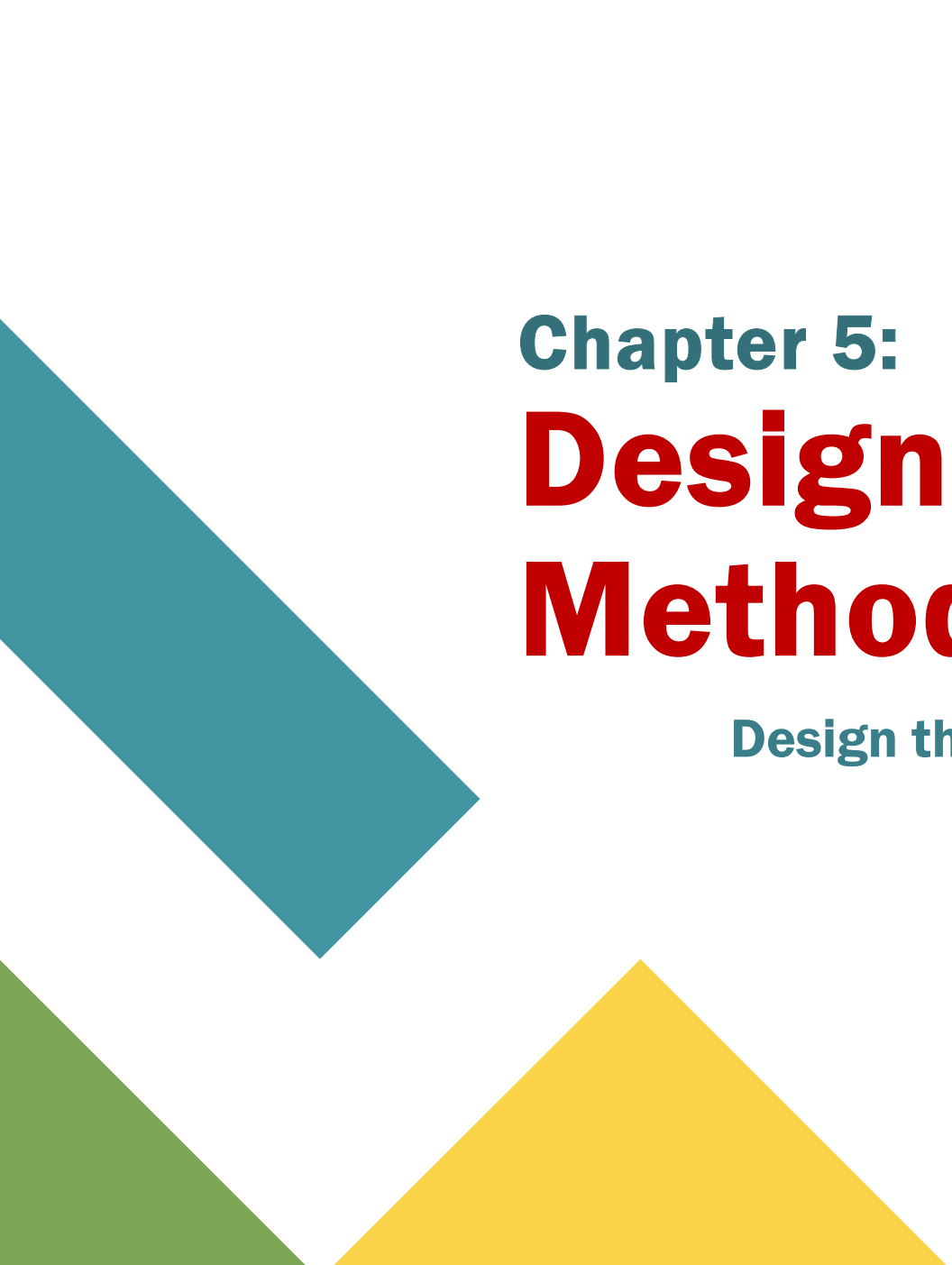




Chapter 5:

Design Thinking Methodology

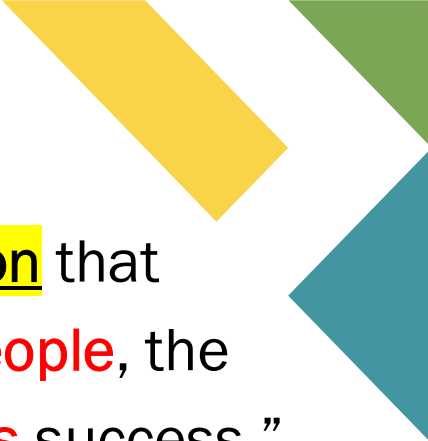
Design thinking from plan to product

Electronic Commerce

Outline

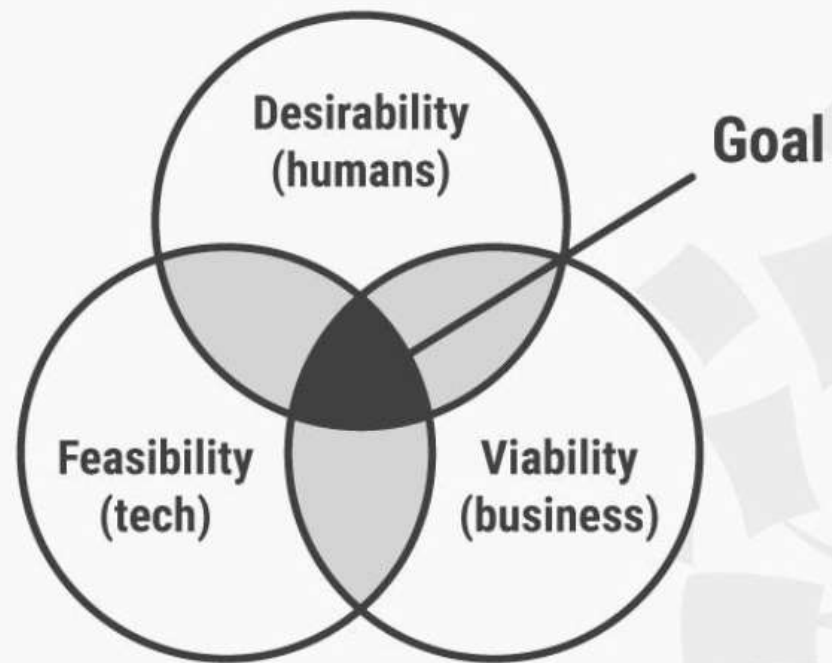
- What is design thinking?
- Hypothesis Testing

What is design thinking?



“Design thinking” is a **human-centered** approach to **innovation** that draws from the designer's toolkit to integrate the **needs of people**, the **possibilities of technology**, and the **requirements for business** success.”

Tim Brown, President and CEO, IDEO



INTERACTION DESIGN
FOUNDATION

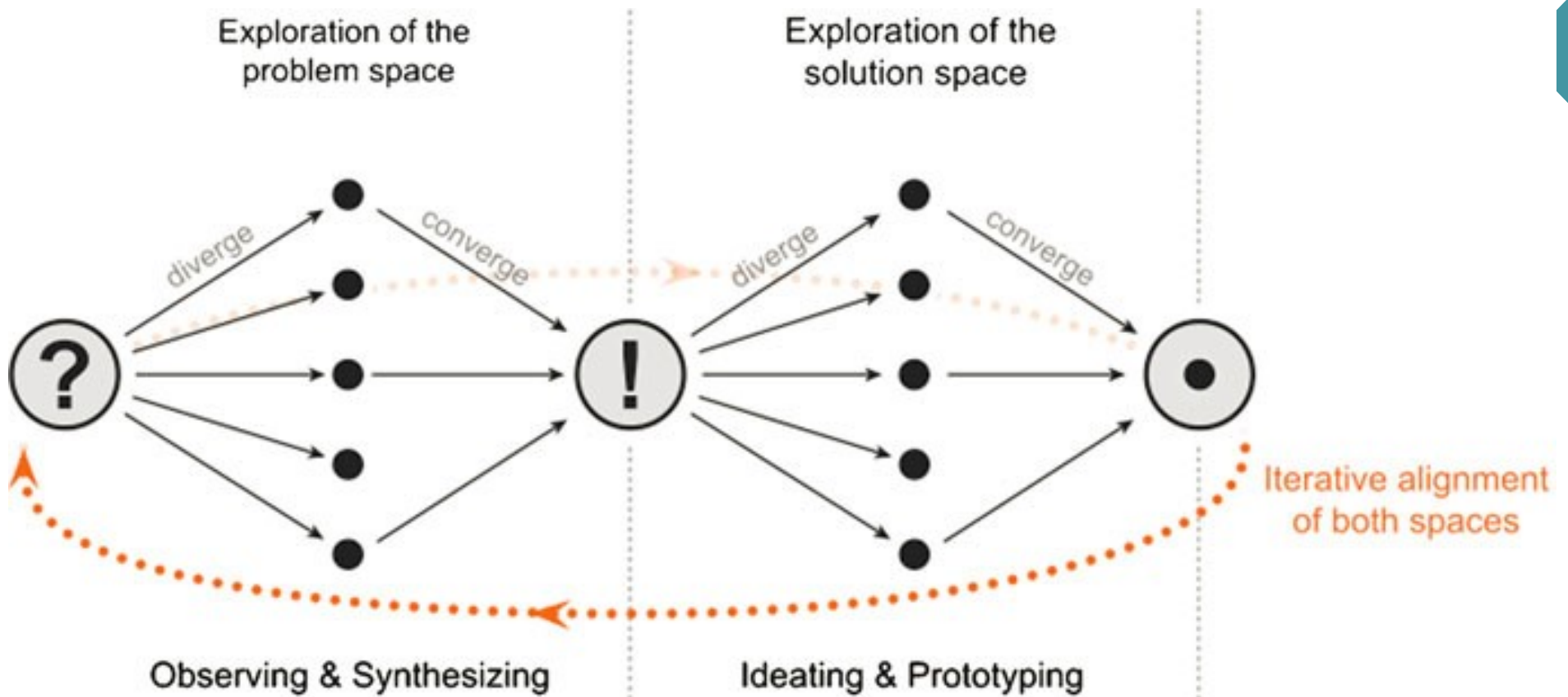
INTERACTION-DESIGN.ORG

What is design thinking?



- A method of focusing innovation on people and designing based on:
 - What people need and want
 - What people like or dislike
 - In regards to production, packaging, marketing, retailing, support, or all of them
- A skill that allows a designer to align what people want with what can be done, and produce a viable business strategy that creates customer value and market opportunity

Spaces of design thinking

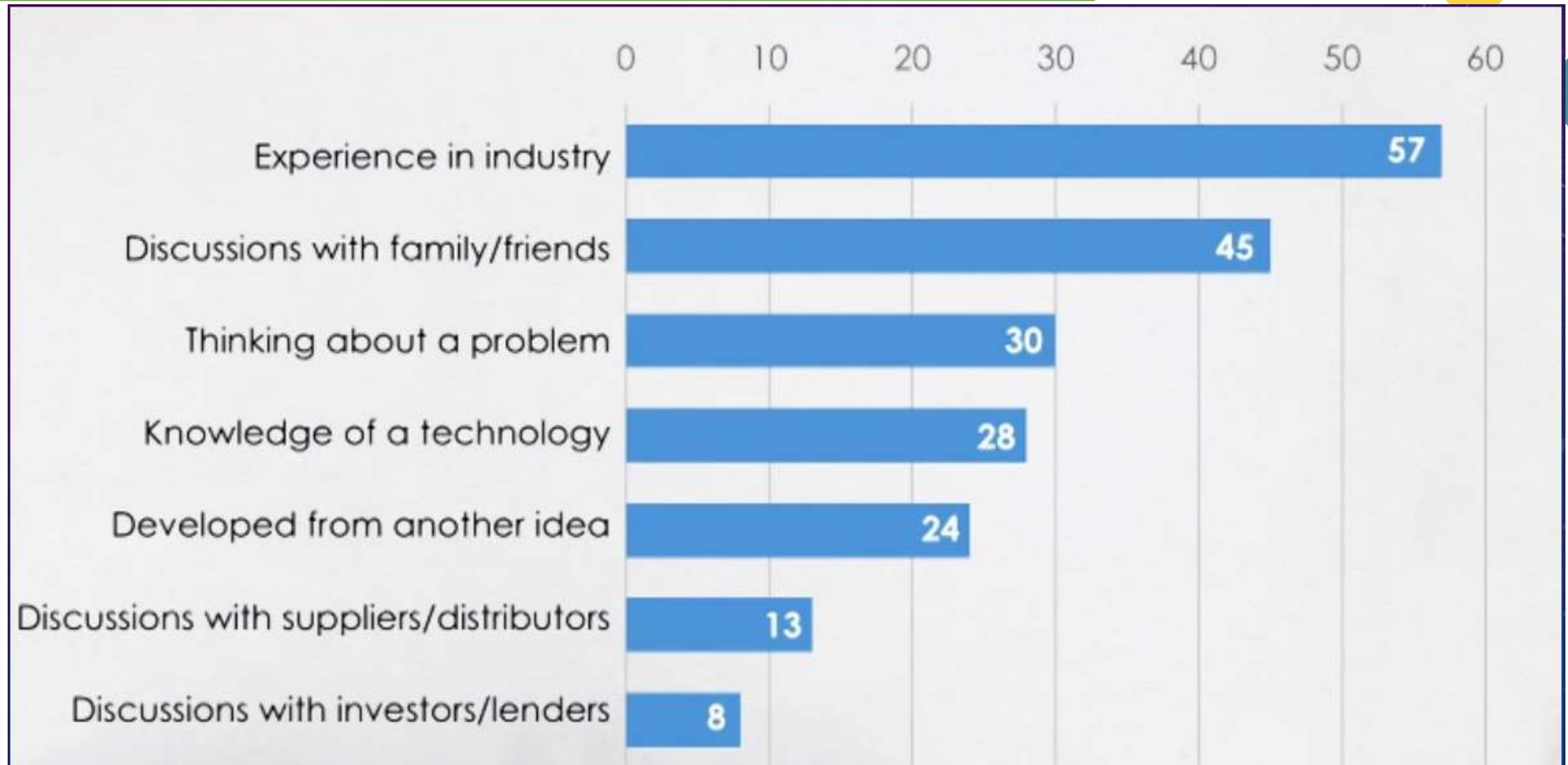


Problem and solution space in Design Thinking adopted to the Stanford Design Thinking model

Observing

- Choose a problem space and target user
- Study current market
 - Current users: likes & dislikes
 - Current techniques
 - History
 - Cost structure
- Understand how things are
- Investigating user needs:
 - Qualitative research techniques: interviews, contextual inquiry, etc.
 - Quantitative research techniques: survey, instrumentation, issues tracking, etc.
 - Investigating existing solutions, competitors, etc.

Where do entrepreneurs get the inspiration for their ideas?



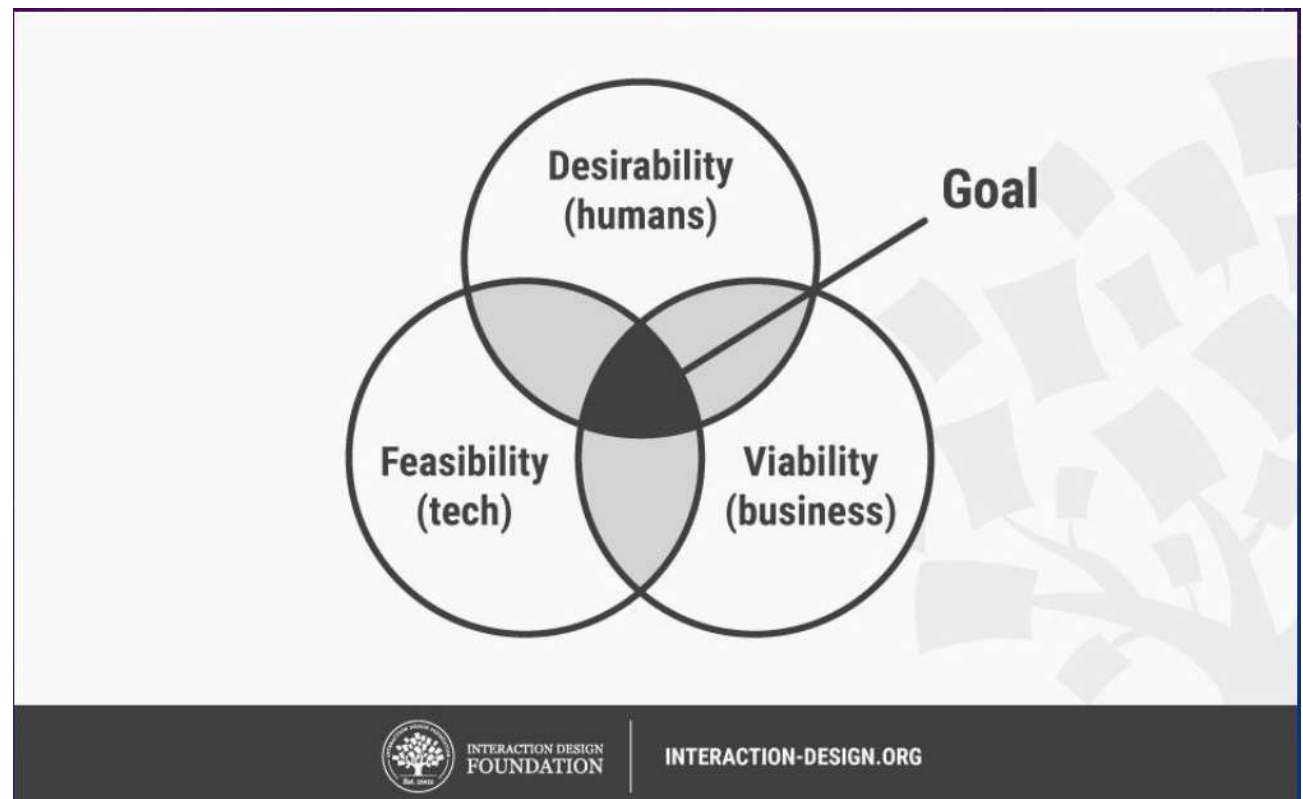
From the Panel Studies asked to 830 companies, ⁸edx.org

Ideating

- Discovery process
 - Starting point: changes in the external environment, such as changes in the regulatory framework, changes in the societal trends or the creation of new technologies
- Creation process
 - Emphasis on the actions entrepreneurs take to create new products for new markets
 - Market creation:
 - High degree of uncertainty, ambiguity , and unpredictability
 - Difficult for one company to create alone

Implementation

- Design experiments
- Test
- Iterate



Hypothesis Testing for Value Creation

Steps

- Hypothesis statement
- Test method
- Test metric & criteria

Example: ScotchFinder startup

- (1) We believe that people will be happy to share their scotch with others
- (2) We will test this by interviewing 100 people outside of a liquor store
- (3) We are right if more than 70% say that they like this idea

Revision

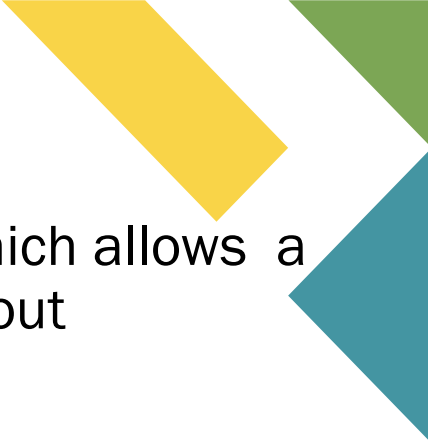
- **Hypothesis statement:** We believe that people will be happy to share their scotch with others
 - Who are the people?
 - Goal: identify our first market segment (a specific group with a specific problem)
 - Does not need to represent the entire population
- **Test method:** We will test this by interviewing 100 people outside of a liquor store
 - What kind of liquor store?
 - The locations
- **Test metric & criteria:** We are right if more than 70% say that they like this idea
 - Actual goal: to learn about the problems that our target group is encountering

Test Card



Name of test	
Hypothesis What assumption will you focus on?	
Test How will you test your hypothesis?	
Metric How will you measure the results?	
Criteria How will you know if your hypothesis is correct?	

Minimum Viable Product - MVP

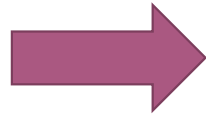


- “A minimum viable product is that version of a new product which allows a team to collect the maximum amount of validated learning about customers with the least effort.” (Eric Ries)
- Then, we put them in front of the people for whom we are trying to create value
- Their responses will tell us if we are on the right track

➔ FAST and CHEAP way to REDUCE our RISK

Scientific Method

GUESS



COMPUTE
CONSEQUENCES

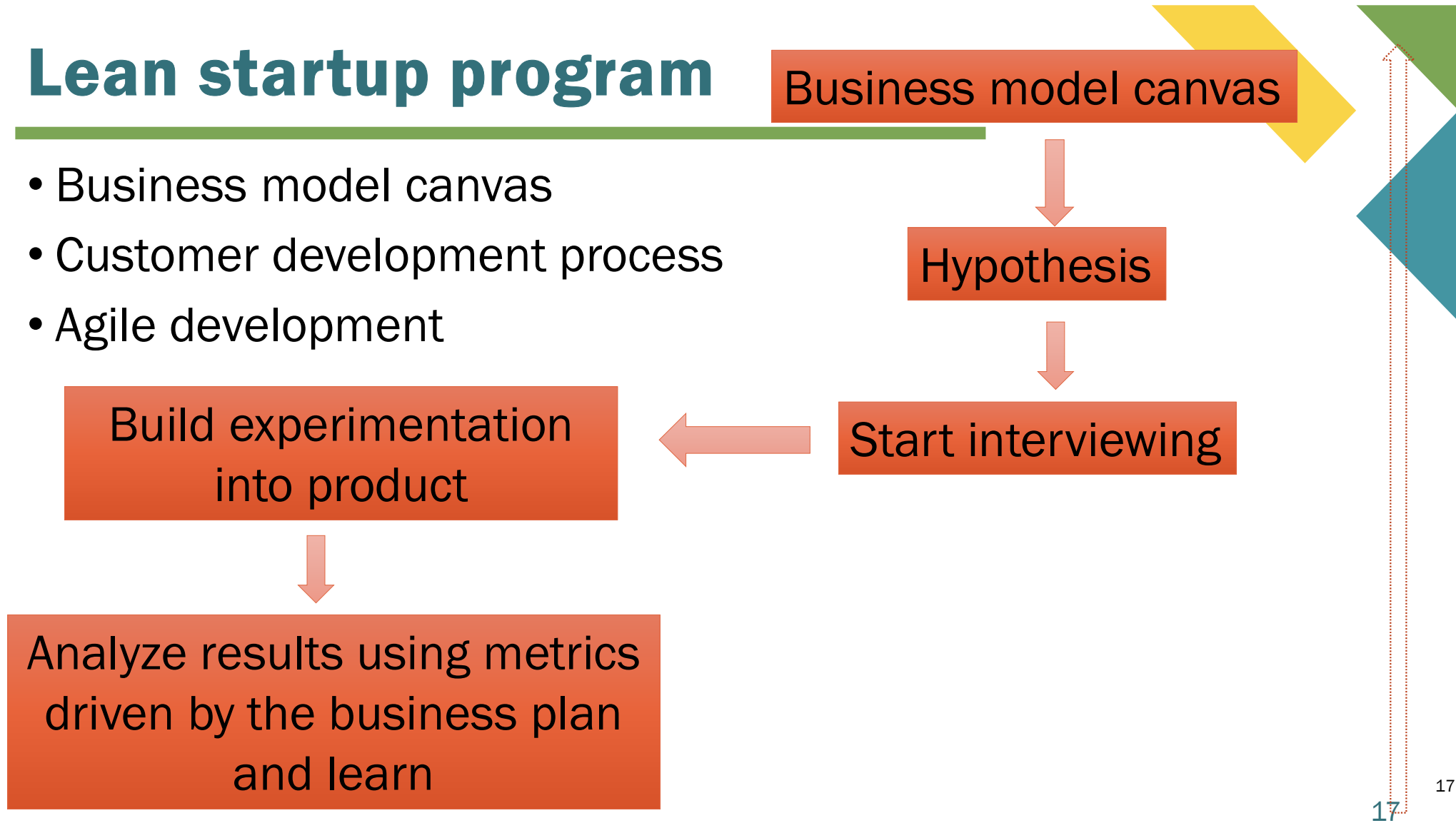


COMPARE
COMPUTATIONS
WITH
EXPERIMENTS

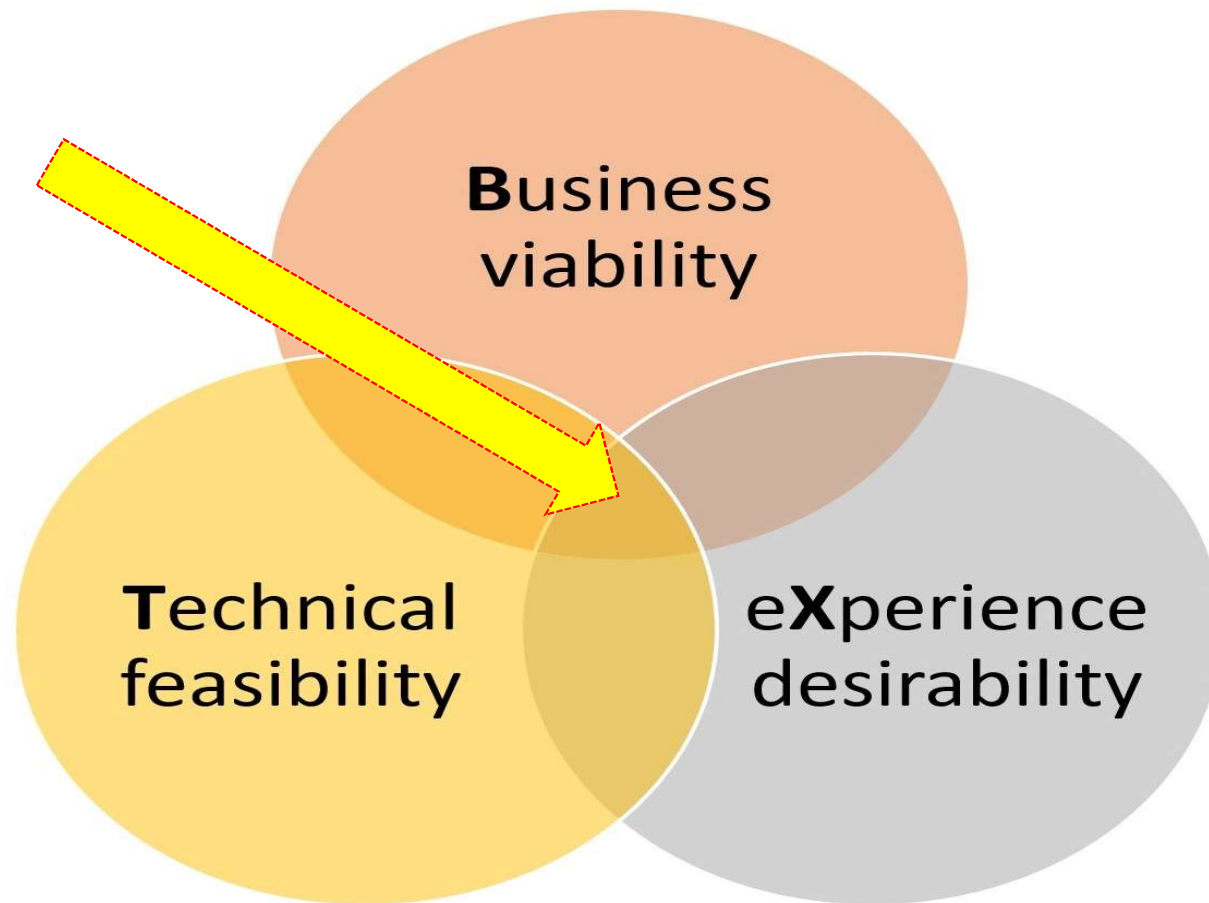
Name of learning card	
Hypothesis What assumption did you have?	
Observation: What did you observe?	
Learning: What did you learn?	
Actions: What are your next steps?	

Lean startup program

- Business model canvas
- Customer development process
- Agile development



Picking winning ideas



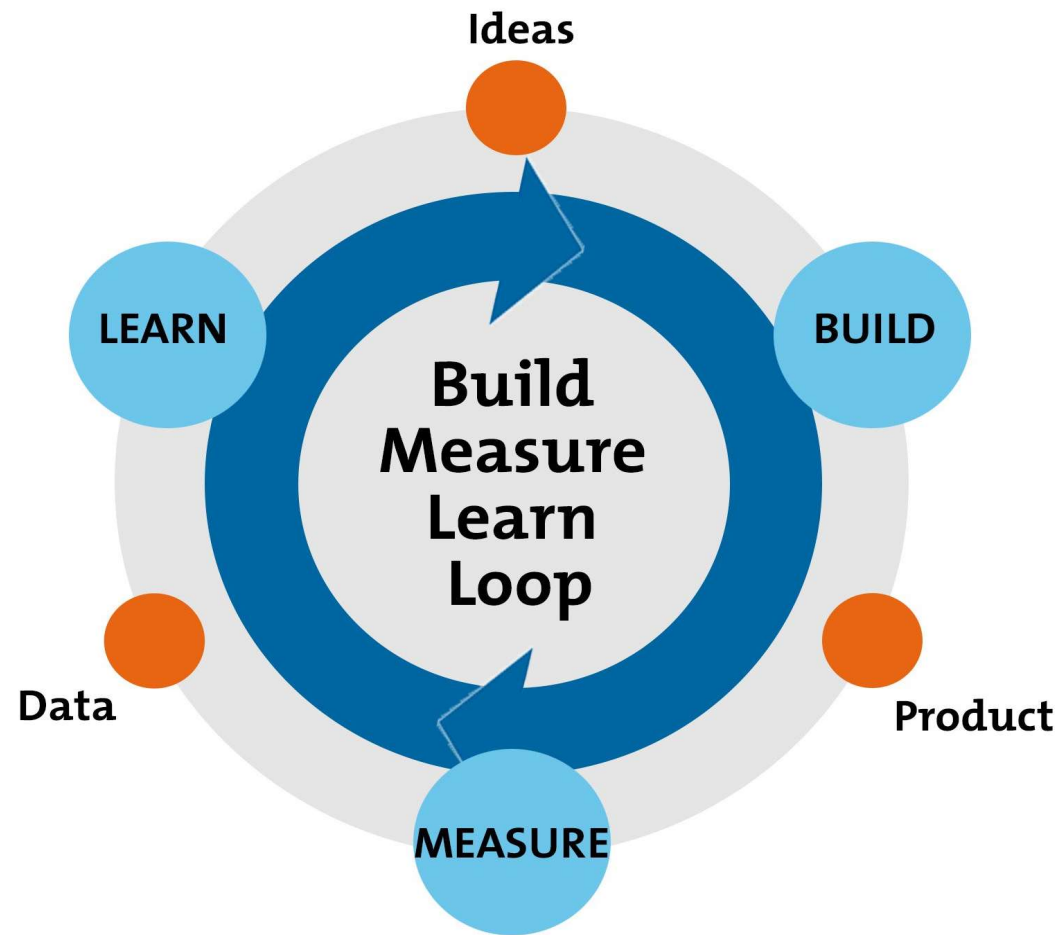
Agile Manifesto

- Agile is a set of values and principles that guide how you should conduct innovation (from 2001)
- Bring four values
 - Individuals and interactions
 - Customer collaboration
 - Responding to change
 - Working versions of products

Principles of Agile Manifesto

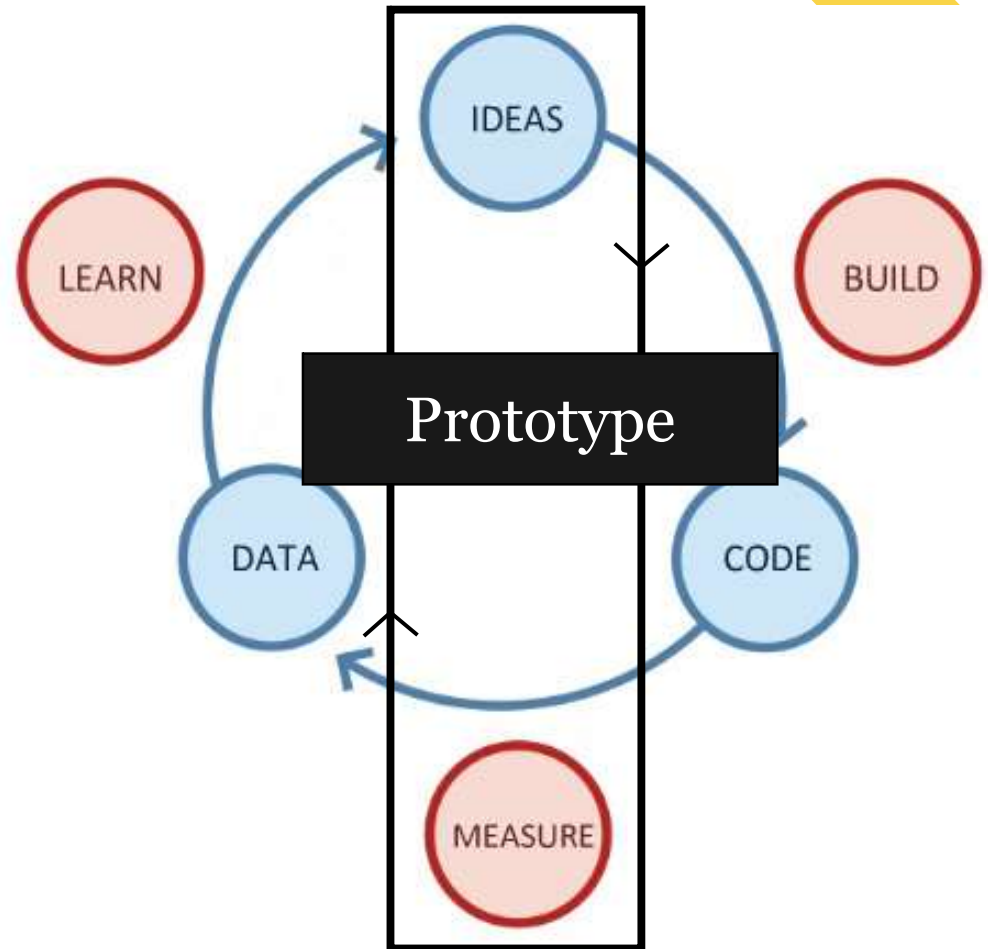
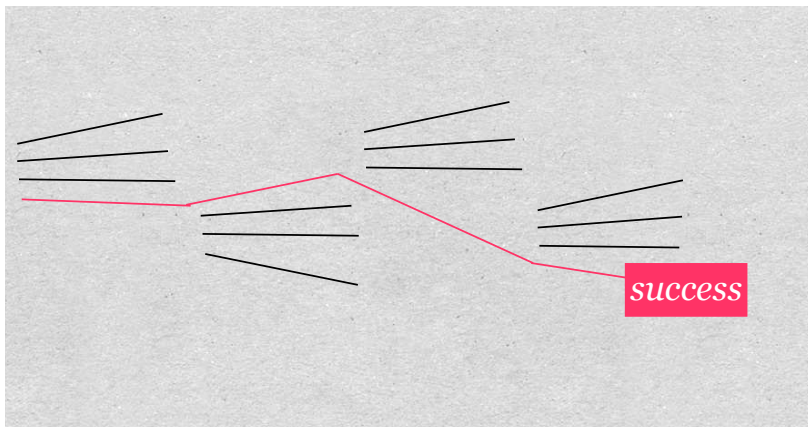
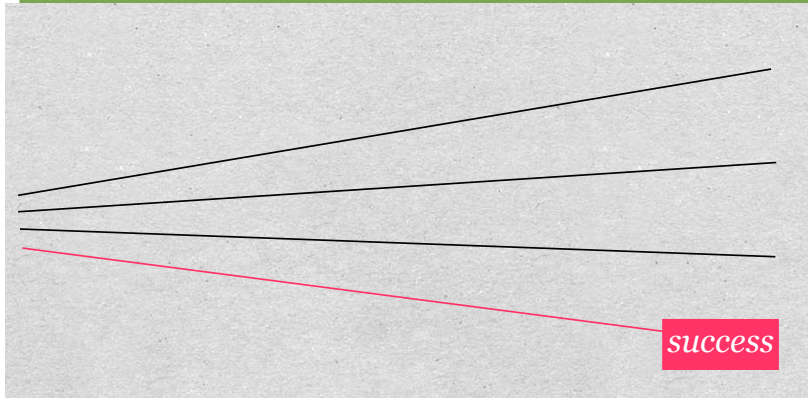


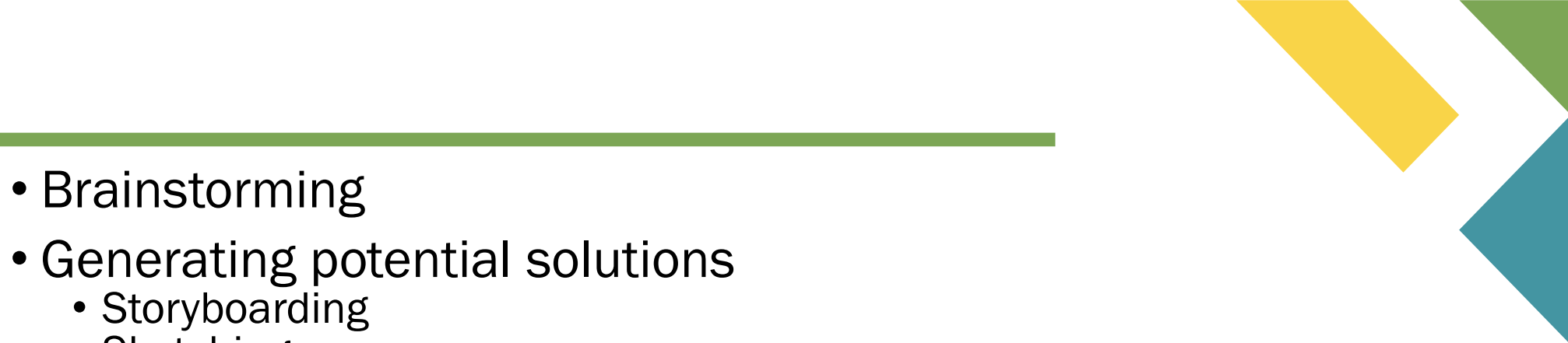
- **Principle 1:** Build – Measure – Learn cycle of design sprint
- **Principle 2:** Measure of improvement
- **Principle 3:** Autonomous team
 - Teamwork spirit
 - Creativity, innovation, and improvement
 - Decision making: Should be empowered to make decision





Prototyping



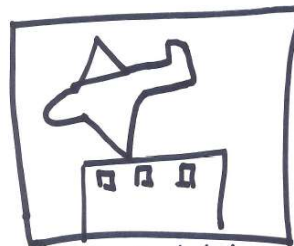
- 
-
- Brainstorming
 - Generating potential solutions
 - Storyboarding
 - Sketching
 - Wireframe
 - Prototype
 - Mockup
 - Information architecture and navigation
 - User testing
 - Tools: Balsamiq, Invision, Figma, Adobe, etc.
 - Building on feedback - Critique

Storyboard

Storyboard for Storyboarding OOC

	A	B	C	D	E
1	http://tinyurl.com/storyboard-slides	Week 1 - Introduction	Week 2 - Storyboarding Tool	Week 3 - Outcomes and Assessment	Week 4 - Activities
2	Outcomes	1. Do the introductions (in terms of people and basic knowledge) and set the scene for the course.	2. Evaluate different tools to create storyboards and select the one that suits you best.	3. Define the learning outcomes and assessment instruments for your course.	4. Design a series of e-tivities (online learning activities) to support the learning outcomes - high level design.
3	Activity 1	Purpose: Introduce yourself to other participants.	Purpose: Evaluate the suitability of Lino as your storyboarding tool.	Purpose: Develop (or review) learning outcome statements.	Purpose 1: Clarify the main purpose of at least one task per unit per week.
4	Activity 2	Purpose: Identify undesirable course features to be avoided.	Purpose: Evaluate the suitability of Popplet as your storyboarding tool.	Purpose: Evaluate your assessment tasks against learning outcome statements.	Purpose 2: Develop spark, task, peer response, technology and timing for at least one e-tivity.
5	Activity 3	Purpose: Describe the course you.	Purpose: Evaluate the suitability of Lino as your storyboarding tool.	Purpose: Evaluate your assessment tasks against learning outcome statements.	Purpose 3: Share and respond to your preferred e-tivities.
6					
7					
8					
9					
10					
11					
12					

Email on phone



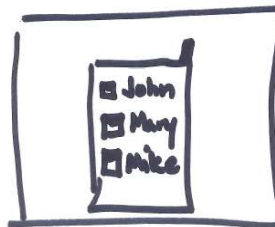
Frank is headed to NYC for client meeting



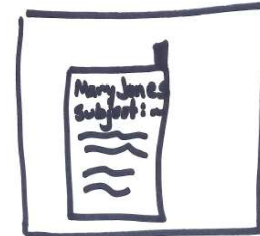
Frank is waiting for his flight at the gate



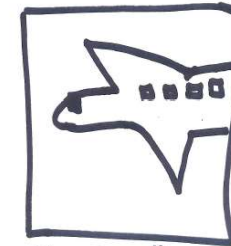
Frank is notified on the phone



Inbox shows up on his phone



Reads & replies to email on his phone



Sleeps well on the plane

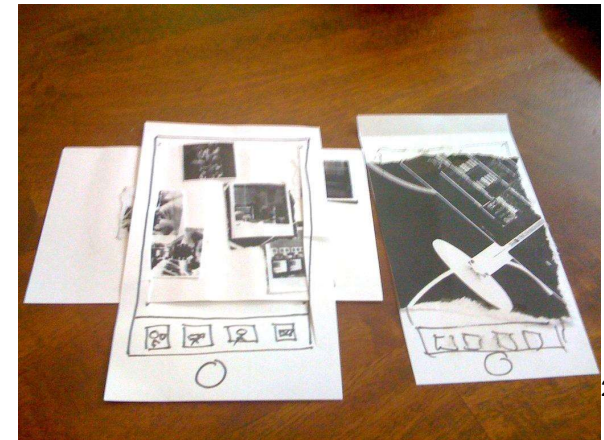
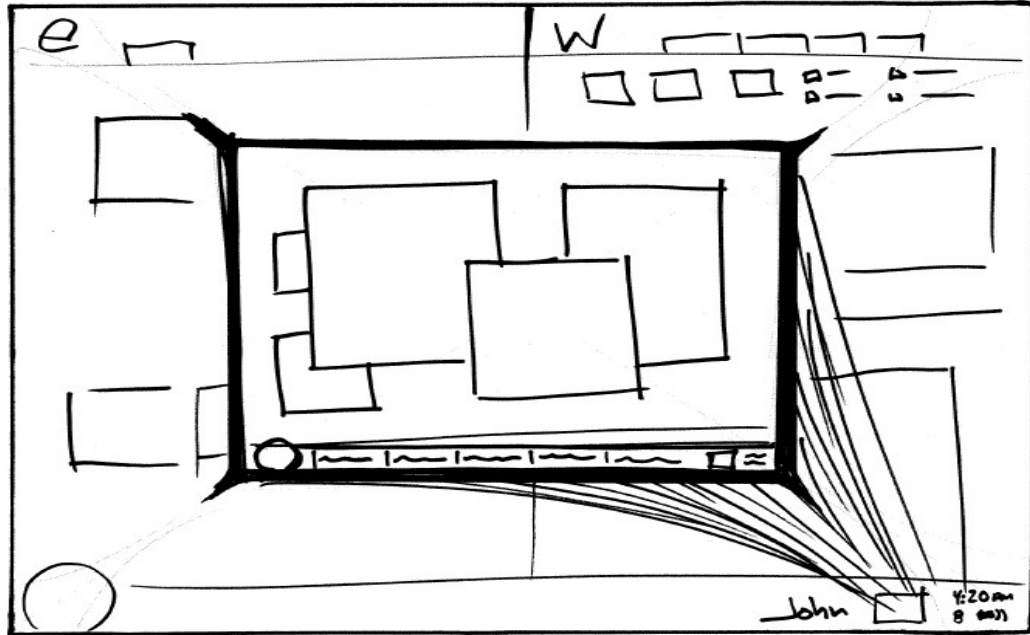
Storyboard



- What do you need? Paper, Pens, A bit of glue, Some post-its, Print-outs of UI-elements (+ scissors)
- Online Tools:
 - <https://wonderunit.com/storyboarder/>
 - <https://www.canva.com/create/storyboards/>
 - Google sheets, draw.io
 - Video makers
- A process of time and interaction
- Prerequisites: personas, scenarios, the scope
- Elements: images, transitions, effects, voiceover, soundtrack

Sketching

- Sketching User Interfaces, User Flow
- When you sketch, you are trying to will work in user interface form.
- Express your ideas quickly and invc experience
- Can sketch on a whiteboard or blar
- Detail is not important, your trying to establish flow



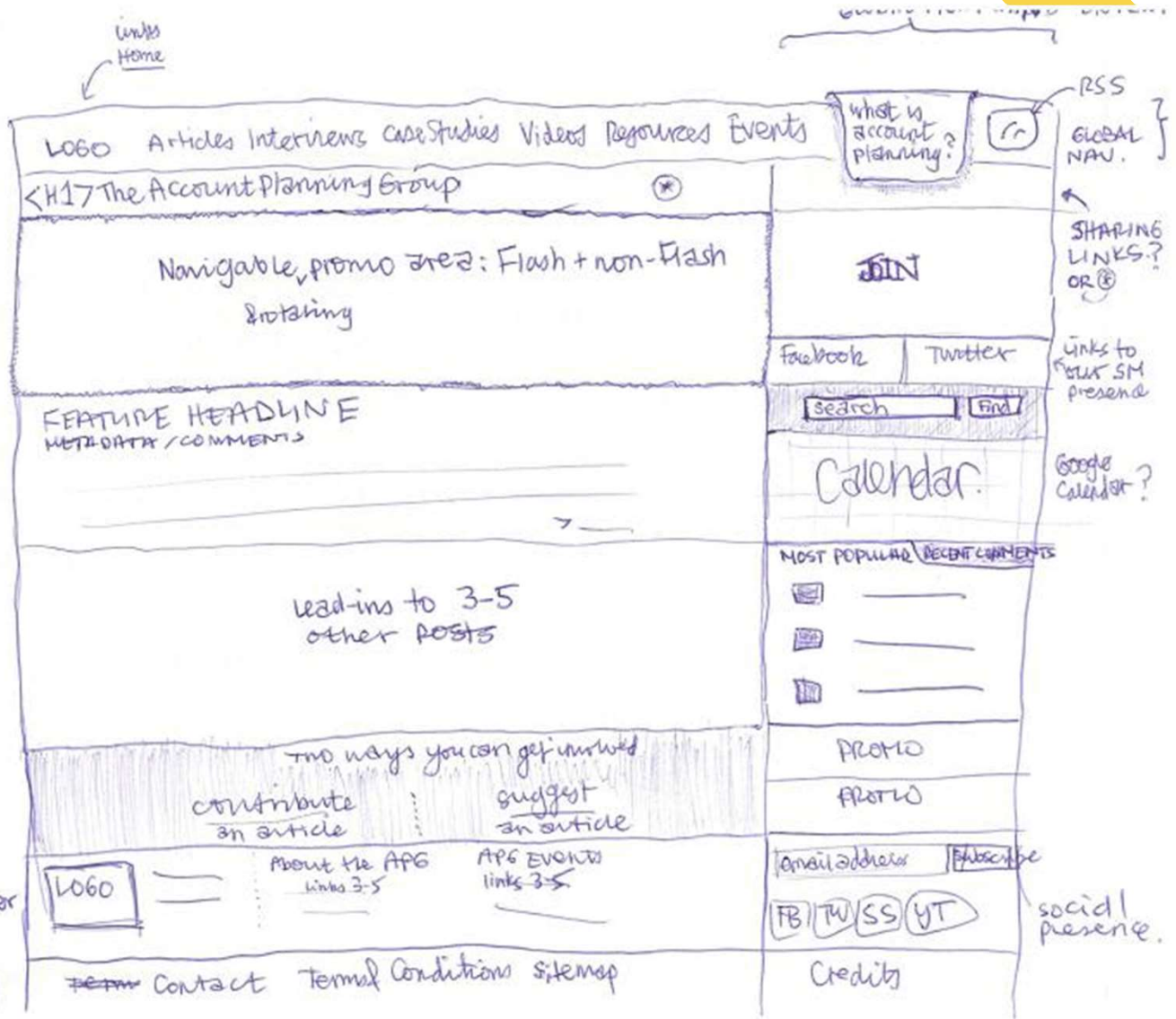
APG
Homepage

CORE THOUGHT:
sparkling ideas

possibly an image?

Global content footer

Global mandatory footer



Steps

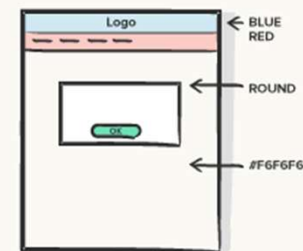
- Define what you need to draw
- Define what keyframes and transitions you should include in your drawing
- Define the starting point.
- Decide what comes next.
- Sketch alternative routes, entries.
- Think about alternative flows.
- Add annotations, notes, details.



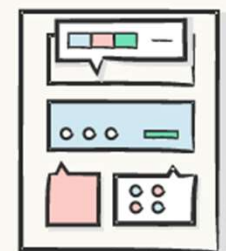
Sketch



Wireframe



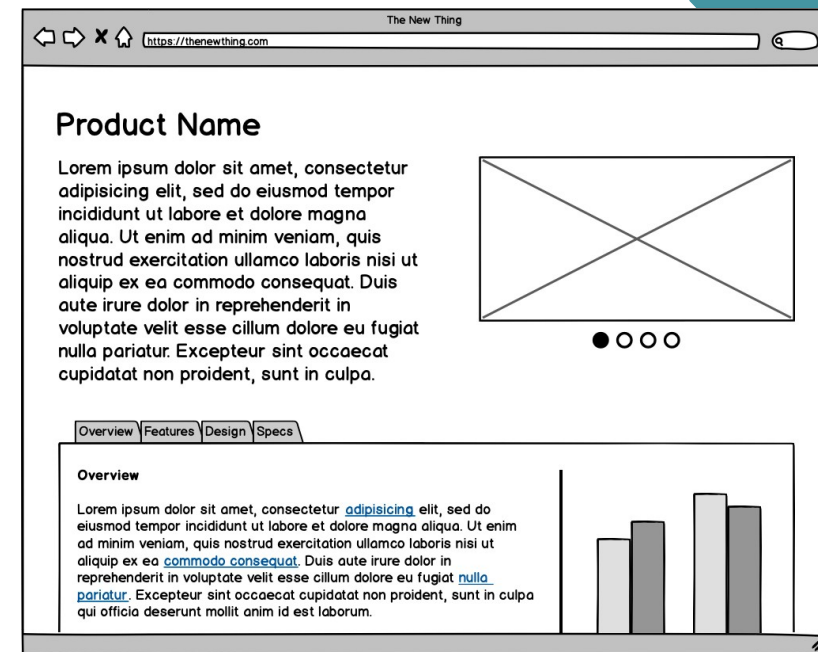
Visual Design

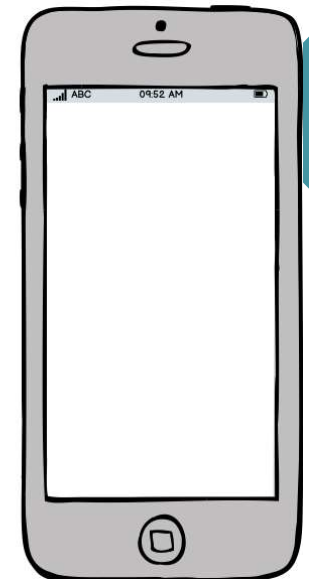
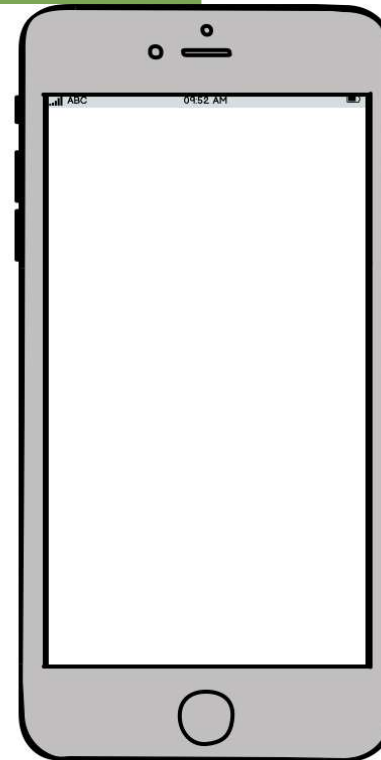
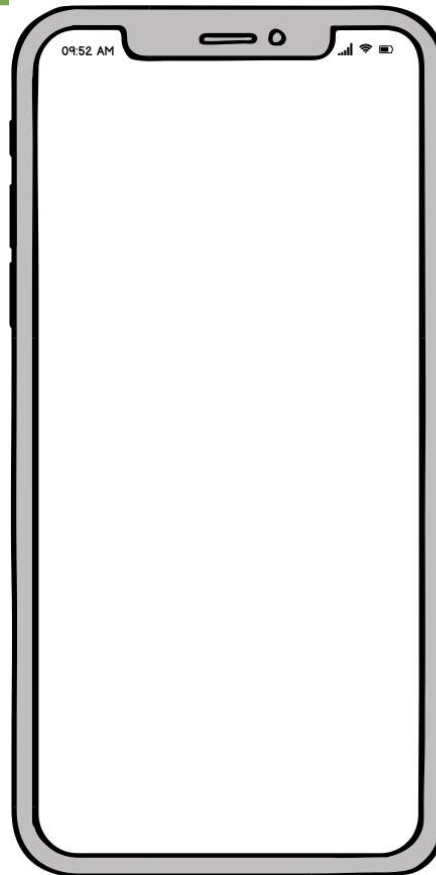
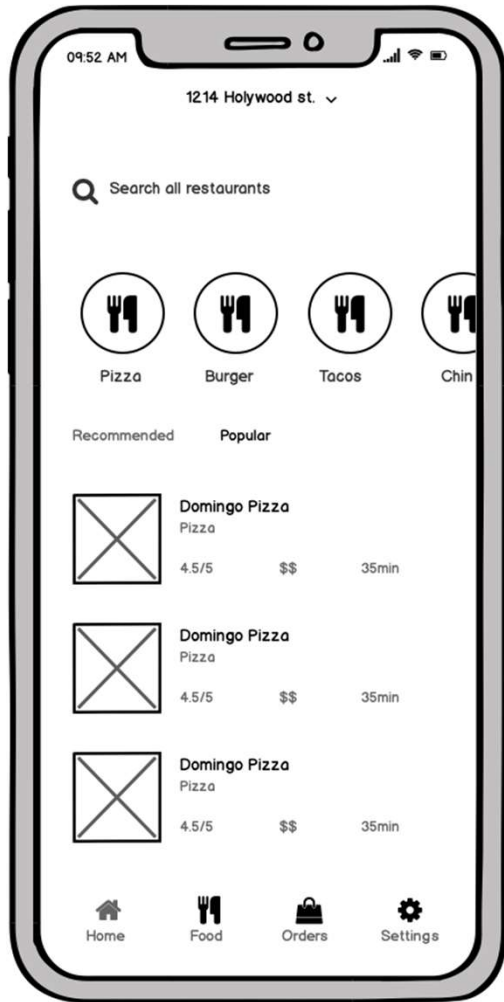


Components

Wireframe

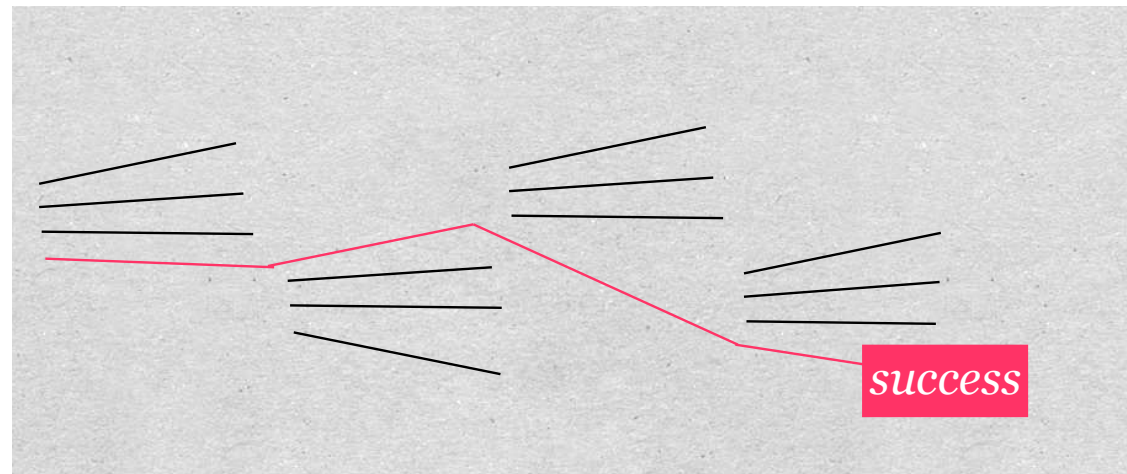
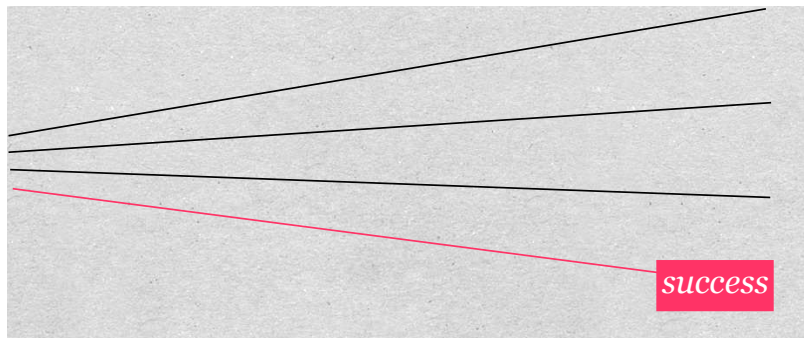
- A wireframe is a schematic or blueprint that is useful for helping you, your programmers and designers think and communicate about the structure of the software or website you're building.
- Wireframes are used early in the development process to establish the basic structure of a page before visual design and content is added.
- Some tools as Sketch, Adobe XD, Axure, Visio, balsamiq, figma, moqups, draw.io





Prototype & Mockup

- Prototype: An interactive version of wireframe
- Mockup: a beautiful version of wireframe

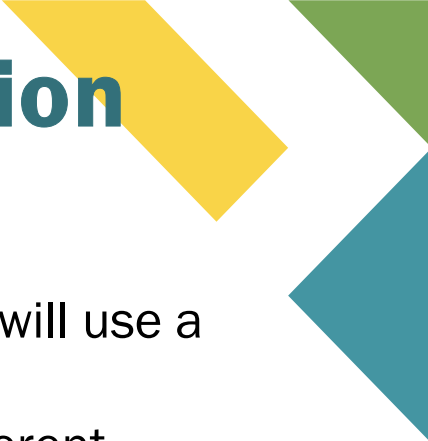


Information Architecture & Navigation



- IA defines site content and functionality, while a website's navigation is a collection of user interface components and patterns that helps visitors find information and functionality.
- **8 principles**
 1. The principle of objects: Content should be treated as a living, breathing thing. It has lifecycles, behaviors, and attributes.
 2. The principle of choices: Less is more. Keep the number of choices to a minimum.
 3. The principle of disclosure: Show a preview of information that will help users understand what kind of information is hidden if they dig deeper.
 4. The principle of exemplars: Show examples of content when describing the content of the categories.

Information Architecture & Navigation



- 8 principles

5. **The principle of front doors:** Assume that at least 50% of users will use a different entry point than the home page.
6. **The principle of multiple classifications:** Offer users several different classification schemes to browse the site's content.
7. **The principle of focused navigation:** Keep navigation simple and never mix different things.
8. **The principle of growth:** Assume that the content on the website will grow. Make sure the website is scalable.

Information Architecture & Navigation

- The principle of front doors



The screenshot displays the Government of Canada website's navigation structure. At the top, the Canadian flag is followed by the text 'Government of Canada' and 'Gouvernement du Canada'. To the right is a search bar labeled 'Search Canada.ca' with a magnifying glass icon, and a link to 'Français'. Below this is a horizontal menu with eight items: 'Jobs', 'Immigration', 'Travel', 'Business', 'Benefits', 'Health', 'Taxes', and 'More services', each with a downward arrow. Under the 'Immigration' menu item, a breadcrumb trail reads 'Home → Immigration and citizenship'. The main content area features the heading 'My immigration or citizenship application' in a large, bold font. Below the heading is a paragraph: 'The basics of applying, such as your account, application status, finding forms, our offices, fees, and medical exams or police checks.' To the right of the text is a photograph of a person's hands typing on a laptop keyboard.

Français

Search Canada.ca

Jobs ▾ Immigration ▾ Travel ▾ Business ▾ Benefits ▾ Health ▾ Taxes ▾ More services ▾

[Home](#) → [Immigration and citizenship](#)

My immigration or citizenship application

The basics of applying, such as your account, application status, finding forms, our offices, fees, and medical exams or police checks.



Information Architecture & Navigation

- The principle of multiple classifications



Information Architecture & Navigation

- The principle of focused navigation

The screenshot displays the Government of Canada website's navigation structure. At the top, there is a header with the Canadian flag, the text 'Government of Canada' and 'Gouvernement du Canada', a search bar labeled 'Search Canada.ca', and a language selector for 'Français'. Below this is a primary navigation menu with links for 'Jobs', 'Immigration', 'Travel', 'Business', 'Benefits', 'Health', 'Taxes', and 'More services', each with a dropdown arrow. A breadcrumb trail reads 'Home → Jobs and the workplace → Find a job'. The main content area is titled 'Student employment' and includes a sub-header 'Job opportunities and Government of Canada hiring programs for students.' accompanied by a photo of a smiling woman. Below this, the page is organized into three columns. The left column, 'Services and information', contains links for 'Work for the government as a student' (with a descriptive paragraph about the FSWEPP) and 'Paid education in the Canadian Armed Forces'. The middle column contains a link for 'Find a student research job in the government' (with a descriptive paragraph about the Research Affiliate Program) and 'Other government opportunities for students'. The right column, 'Most requested', lists three links: 'Login to FSWEPP account', 'How to write a good resume', and 'Student rates of pay in the government'. At the bottom right, a 'Contributors' section lists 'Employment and Social Development'.

Government of Canada / **Gouvernement du Canada** Français

Search Canada.ca

[Jobs](#) ▾ [Immigration](#) ▾ [Travel](#) ▾ [Business](#) ▾ [Benefits](#) ▾ [Health](#) ▾ [Taxes](#) ▾ [More services](#) ▾

[Home](#) → [Jobs and the workplace](#) → [Find a job](#)

Student employment

Job opportunities and Government of Canada hiring programs for students.



Services and information

Work for the government as a student
Apply to be considered for full or part-time job opportunities across the country through the Federal Student Work Experience Program (FSWEP).

Find a student research job in the government
Conduct innovative research in your field of graduate or undergraduate study and gain experience with federal departments and agencies through the Research Affiliate Program.

Paid education in the Canadian Armed Forces

Other government opportunities for students

Most requested

- [Login to FSWEP account](#)
- [How to write a good resume](#)
- [Student rates of pay in the government](#)

Contributors

- [Employment and Social Development](#)

Information Architecture & Navigation

- The principle of growth – passing example

Canadian citizenship

Apply to become a Canadian citizen, prepare for the citizenship test and find out how you can resume or give up your Canadian citizenship.



Apply for citizenship

Eligibility requirements, application steps and processing times for Canadian citizenship

Prepare for the citizenship test

Study material and details about who has to take this test, how to get a test date and what happens after you take the test

Get proof of citizenship

Applications and processing times for a citizenship certificate and a list of documents accepted as proof of Canadian citizenship

Resume citizenship

Steps to take if you are a former citizen who willingly gave up Canadian citizenship and want to become a Canadian citizen again

Give up citizenship

Steps to take to renounce your Canadian citizenship if, for example, you want to become a citizen of a country that does not allow dual citizenship

Celebrate being Canadian

Citizenship ceremonies and activities, Canadian history and contests and awards

Prepare for the citizenship ceremony

How to prepare for your special day, what happens during the ceremony, including getting your citizenship certificate, and what to do after the ceremony.

Information Architecture & Navigation

- The principle of growth – failing example

Government of Canada / Gouvernement du Canada

Search Canada.ca

Jobs Immigration Travel Business Benefits Health Taxes More services

Home → Travel → Destinations

France

Register Travel insurance Destinations

Last updated: September 16, 2016 10:50 ET
Still valid: September 22, 2016 11:40 ET
Latest updates: An editorial change was made.

Print format

Risk level (s) Safety and security Entry/exit requirements Health Laws and culture Natural disasters and climate Assistance

Risk level(s)

France - Exercise a high degree of caution

There is no nationwide advisory in effect for France. However, you should exercise a high degree of caution due to the current elevated threat of

Value of IA

- Value for the user
 - 4 main types of needs
 - **Known-item seeking:** Users will come to the website to search for something desirable and known.
 - **Exploratory seeking:** Users will come to the website looking for inspiration. They're looking for something desirable but not sure what exactly.
 - **Exhaustive research:** Users are in a process of an extensive research. They want to find as much information as possible.
 - **Re-finding:** A user needs a desired items again and are trying to find it.

Value of IA

- Employee Productivity
- Sales And Reputation
- Acquiring New Members
- Reducing Marketing Costs
- Reputation And SEO Ranking
- Reducing The Cost Of Live Help And Support Documentation

Information Architecture

- Key processes

- ☐ Define the company goals
- ☐ Define the user's goals
- ☐ Analyze competitors
- ☐ Define content

- ☐ Steps

- ☐ Step 1: Group and label the content
- ☐ Step 2: Define navigation and create site map
- ☐ Step 3: User testing

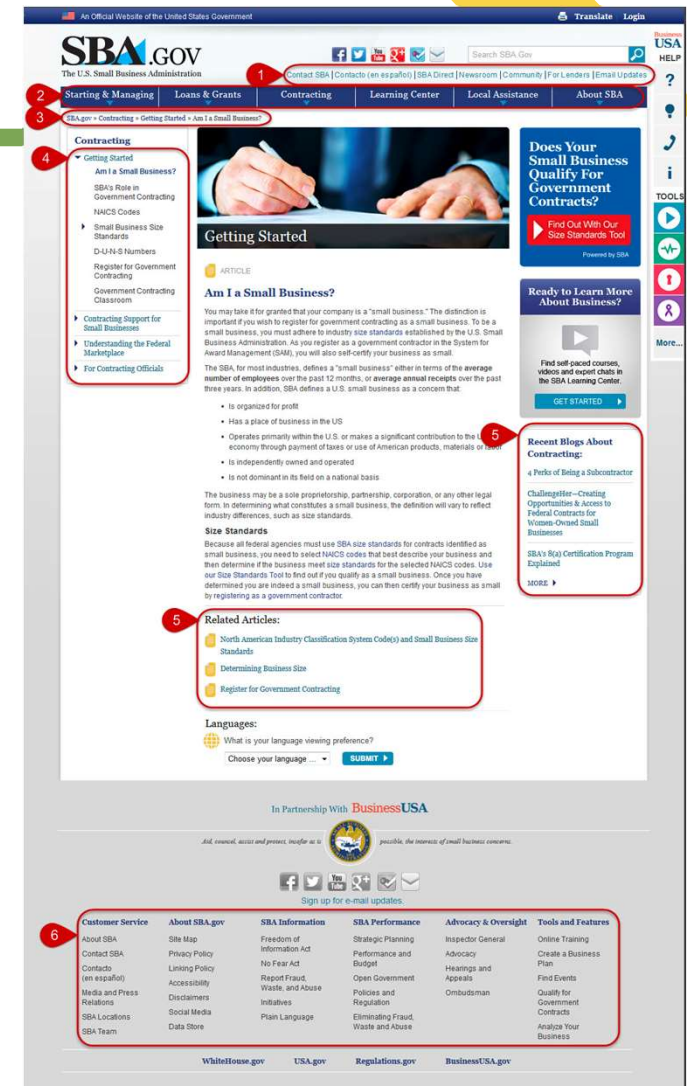
Navigation

- A website or mobile app's navigation is a collection of user interface components.
- Help users find information and functionality, and encourage them to take desirable actions.
- Navigation components include global navigation, local navigation, utility navigation, breadcrumbs, filters, facets, related links, footers, fat footers, and so on.

Navigation

- Some examples of navigation components:

- 1) utility navigation;
- 2) global navigation;
- 3) breadcrumb trail;
- 4) local navigation;
- 5) related links (in this case, articles and blog posts);
- 6) fat footer. (www.sba.gov)

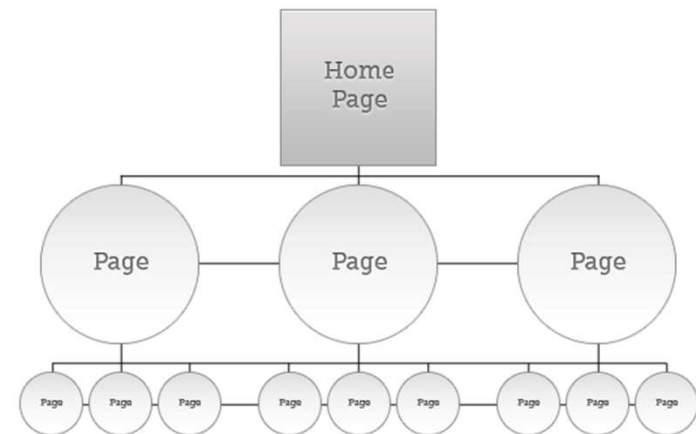
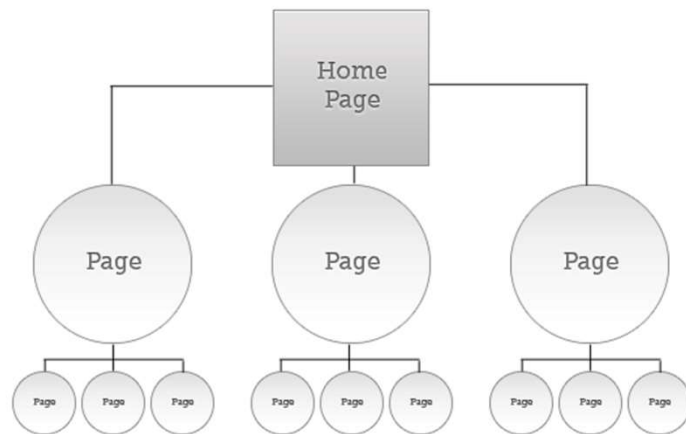


Content organization models

- Single page model: This model works best for landing pages.
- Flat model:
 - All the pages are equally important, and they are put at the same level of hierarchy in the navigation system.
 - This model works best for small websites.



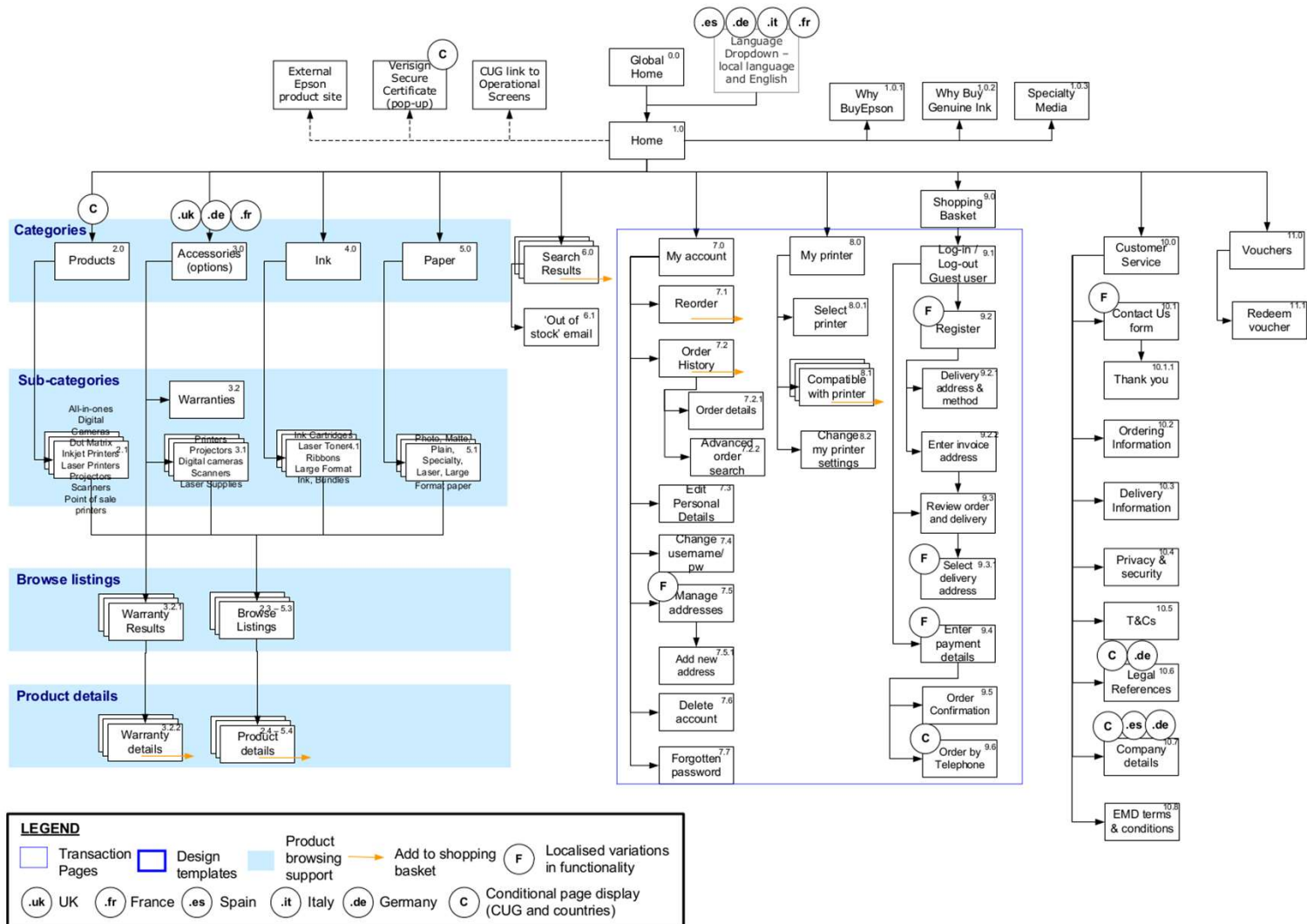
- Hierarchical model



Sitemaps



- Application maps/site maps
 - A document describes how the components you've designed are incorporated in the existing system.
 - Demonstrates the information hierarchy architecture of the entire design, surfaces critical pathways through which the user can navigate
 - Offers a comprehensive view of all the work that needs to be implemented.
- Interaction design
 - Designers may be expected to create documentation regarding how the user interface should be built and interaction between modules



User Testing

- To help understand how people are using your designs
- From third parties
 - A quantitative measurement
 - Tell you what users are doing with your released software
 - For example: Google analytics, Microsoft's the Software Quality Metrics (SQM)
- A/B testing



ccdsign.com

All Web Site Data ▾



Search reports and help



CUSTOMIZATION

Reports



REAL-TIME



AUDIENCE

Overview

Active Users

Lifetime Value BETA

Cohort Analysis BETA

User Explorer

▶ Demographics

▶ Interests

▶ Geo

▶ Behavior

▶ Technology

▶ Mobile

▶ Custom

▶ Benchmarking



ADMIN

Users Flow

May 5, 2017 - May 11, 2017 ▾

Level of Detail ▾ Export ▾



All Users
100.00% Sessions



+ Add Segment



This report was generated on 5/12/17 at 4:05:15 PM - Refresh Report

50



Visual Elements

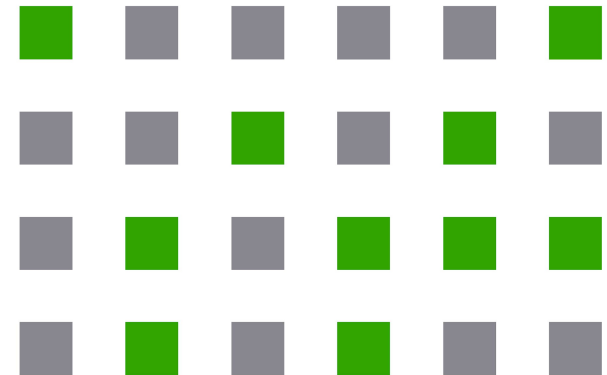
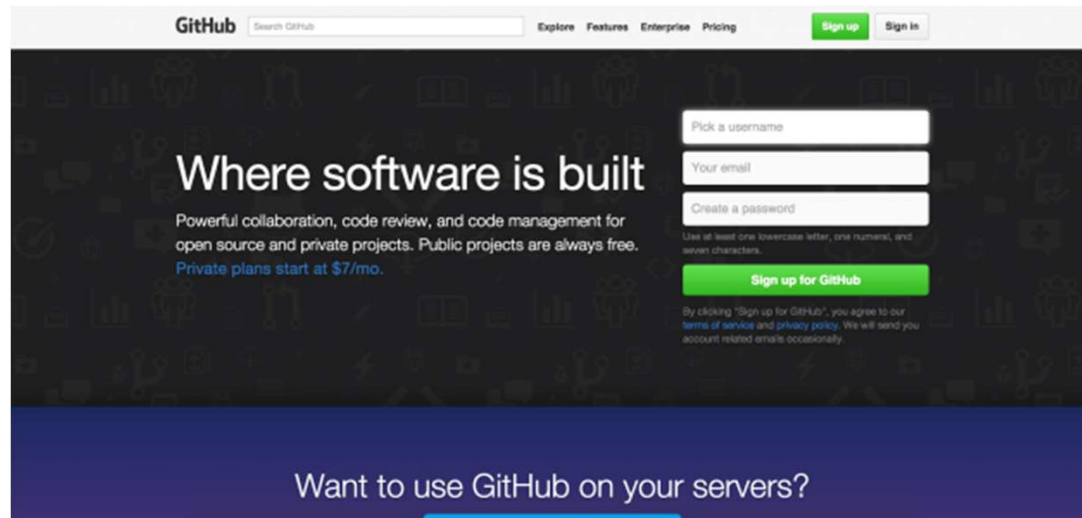
Fundamentals of Visual Design



- Sensation, Perception, Cognition
 - Sensation informs us about our context in the world.
 - Perception is the process of interpreting our sensory input
 - Cognition is the process that assists us in remembering, knowing, judging, and solving problems.
- Components of Gestalt in Design
 - Similarity
 - Proximity
 - Common Region
 - Continuation
 - Closure
 - Figure and Ground
 - Others: Focal point, Symmetry & order, Past experience or Habit, Parall

Similarity

- The principle of similarity states that when things appear to be similar to each other, we group them together. And we also tend to think they have the same function.
- The squares here are all equally spaced and the same size, but we automatically group them by color, even though there's no rhyme or reason to their placement.

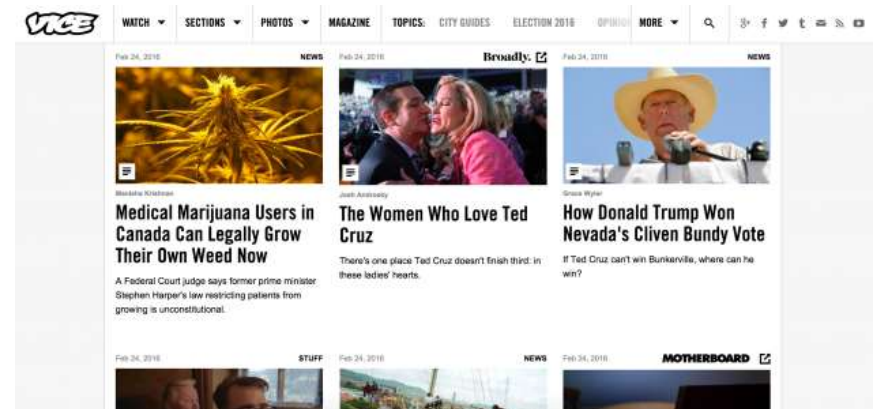
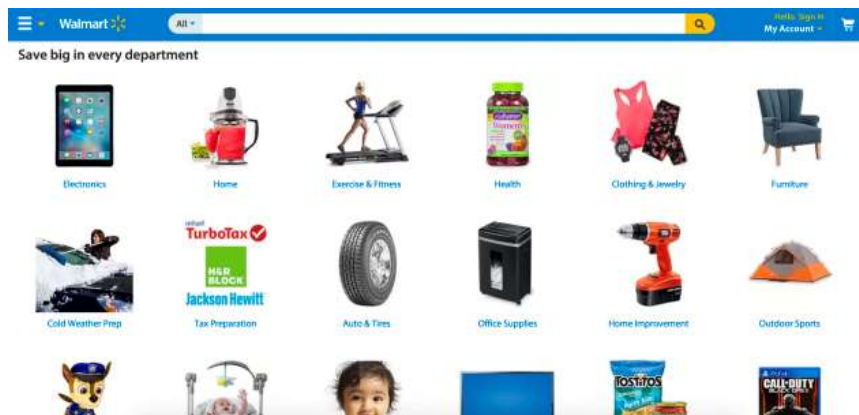
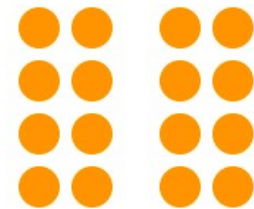
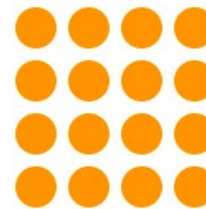


Proximity

- The principle of proximity states that things that are close together appear to be more related than things that are spaced farther apart.

Email

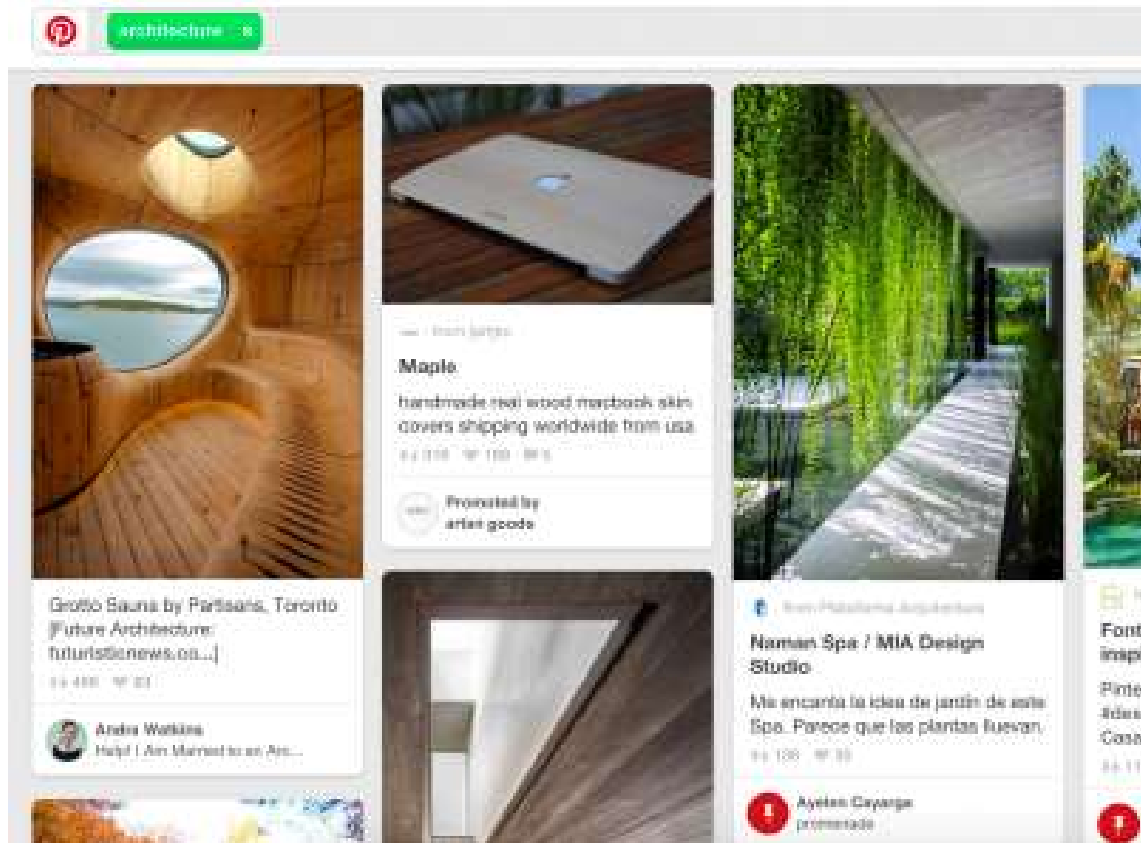
James@stewart.connect



Common region



- The principle of common region is highly related to proximity. It states that when objects are located within the same closed region, we perceive them as being grouped together.
- *A good Common Region example would be the card UI pattern; a well defined rectangular space with different bits of information presented as one. Banners and tables are good examples as well.*



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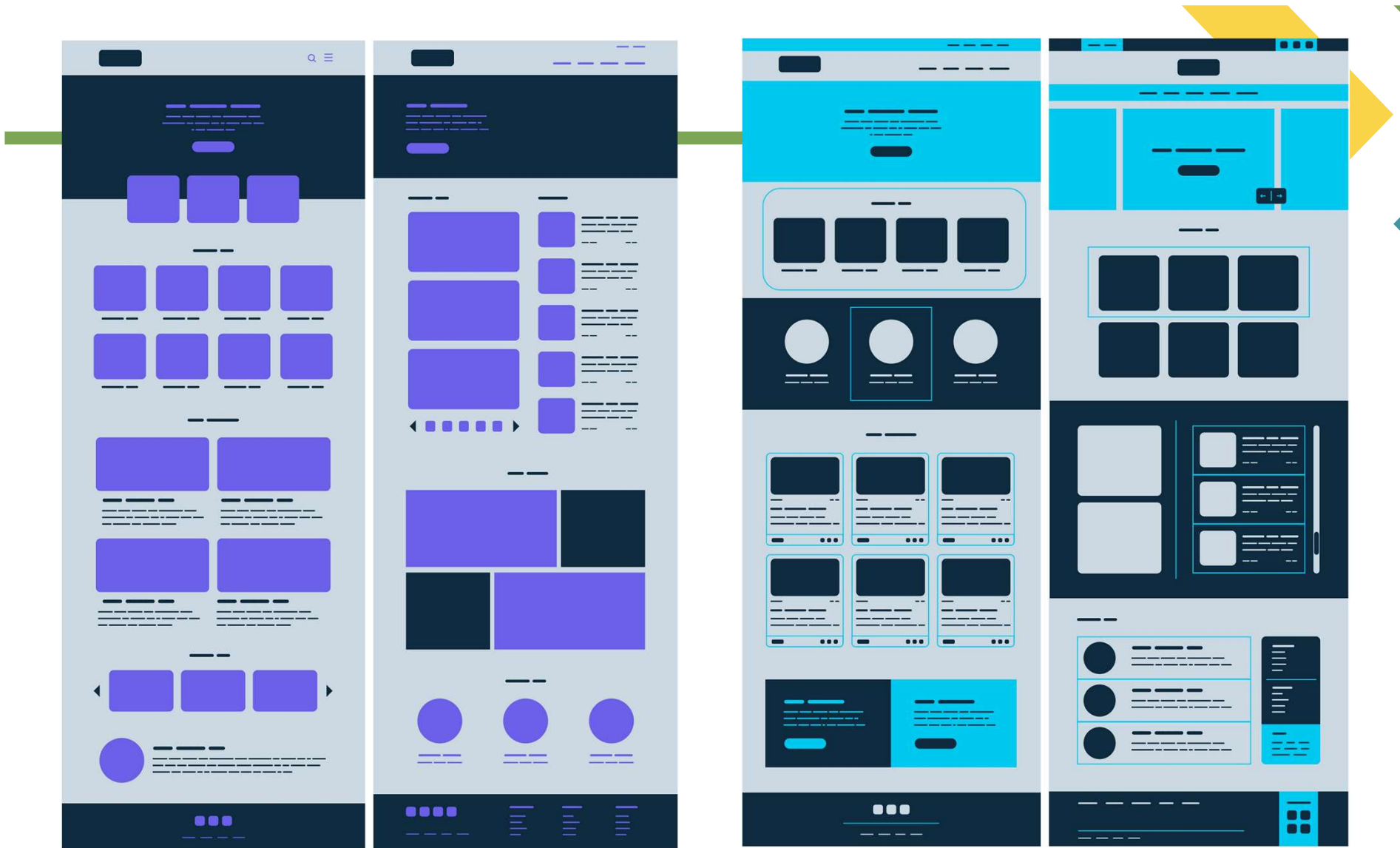
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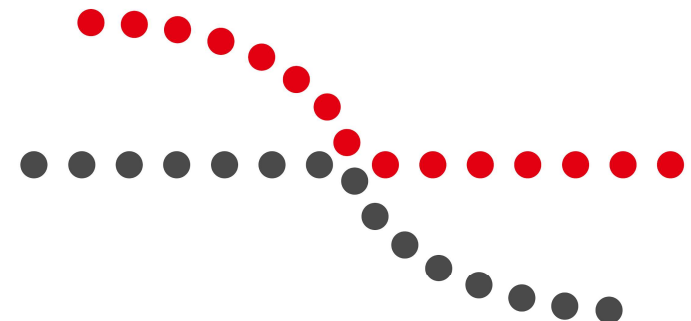
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Continuation

- The law of continuity posits that the human eye will follow the smoothest path when viewing lines, regardless of how the lines were actually drawn.
- The eye tends to want to follow the straight line from one end of this figure to the other, and the curved line from the top to the bottom, even when the lines change color midway through.
- Amazon uses continuity to communicate that each of the products below is similar and related to each other.



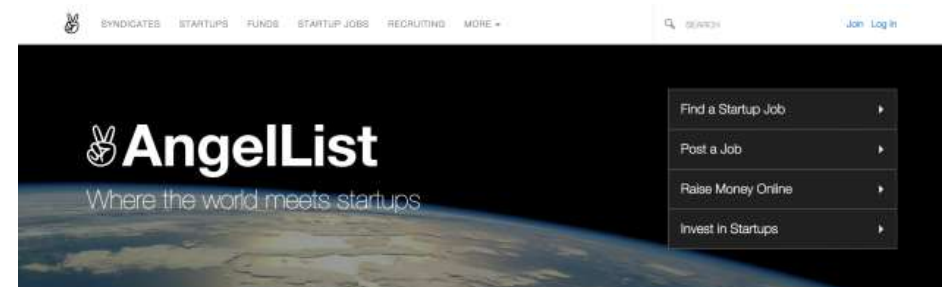
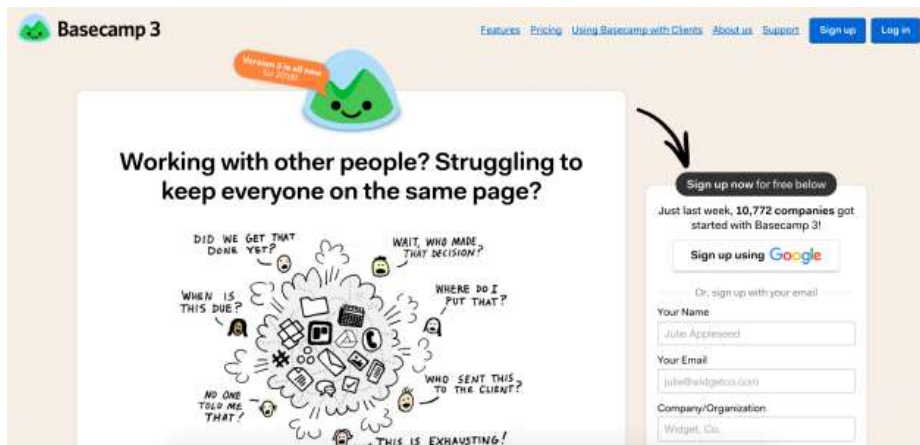
Closure

- The principle of closure states that when we look at a complex arrangement of visual elements, we tend to look for a single, recognizable pattern.
- When you look at the image above you most likely see a zebra even though the image is just a collection of black shapes. Your mind fills in the missing information to create a recognizable pattern based on your experience.



Figure-ground

- The figure-ground principle states that people instinctively perceive objects as either being in the foreground or the background.





Putting a different ending

World famous and oscar winning movie Shape of the water loved by everyone, but what if this movie had different ending. Check out!

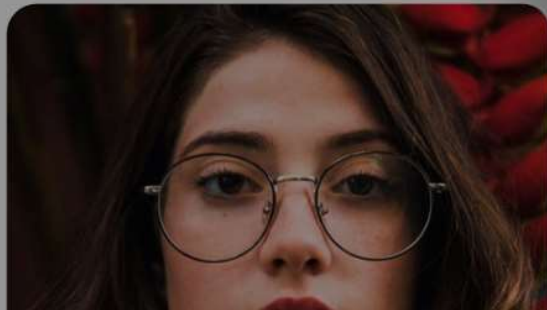


Far Most Gone

Trekking a journey from becoming a beginner to the professional. A journey toward freedom of self expression.



You can focus on this message because of this law.

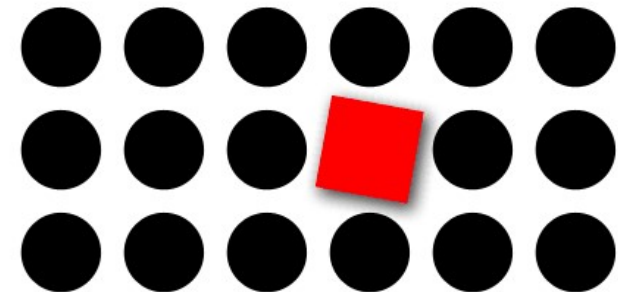
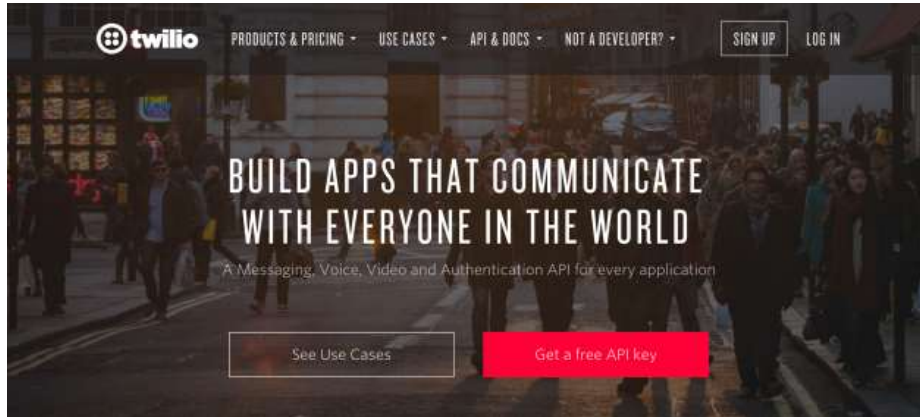


Hidden chambers of love

Being loved is the best experience you can have

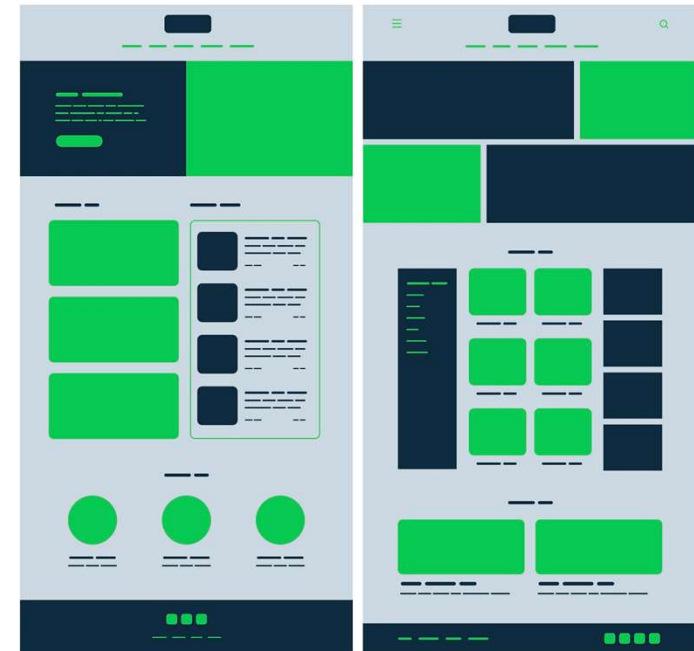
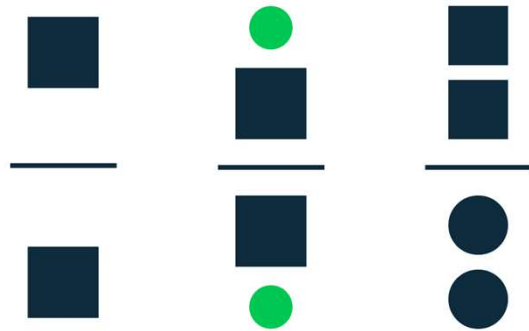
Focal point

- The focal point principle states that whatever stands out visually will capture and hold the viewer's attention first.



Symmetry

- Symmetrical elements tend to be perceived as belonging together regardless of their distance, giving us a feeling of solidity and order.
- *It's good to use Symmetry for portfolios, galleries, product displays, listings, navigation, banners, and any content-heavy page.*



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